

Preserving Sanskrit: Dataset Curation and AI-Based Fine-tuned Model for Grammar Error Correction

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Abstract

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Sanskrit is a living archive of India's civilisational memory. As the origin of most Indian languages and the lingua franca of pre-modern sciences including mathematics, astronomy, and medicine, Sanskrit has carried millennia of philosophical, literary, and scientific knowledge across generations. It preserves the foundational stories of Indian civilisation. Its highly structured grammar and phonetic precision have been linked to cognitive development, and it remains deeply compatible with formal computational systems. To preserve Sanskrit is to preserve access to one of humanity's richest intellectual and cultural traditions, and enabling intelligent AI tools for Sanskrit is therefore not a narrow technical exercise, but an act of cultural stewardship.

Sanskrit texts are being processed using applications like machine translation, post-processing of OCR results, and morphological analysis. However, Grammar Error Correction (GEC) remains unexplored, despite its foundational role in enabling intelligent language learning applications and automated essay scoring systems.

This paper presents the first publicly available dataset for Sanskrit grammar error correction. SanskritGEC_V1.0, a manually annotated dataset of 855 sentence pairs comprising correct and erroneous sentences, covering seven error types has been curated. Additionally, a large synthetic dataset, SanskritGEC_V2.0, comprising 1,03,798 sentence pairs, is generated by employing linguistically inspired error generation methods. Further, the pretrained ByT5-Sanskrit model is fine-tuned via Low-Rank Adaptation for developing a novel Grammar Error Correction system.

The proposed model attained a BLEU score of 83.73 and a Character Error Rate of 10.59%, thus setting the first benchmark for Sanskrit grammatical error correction. To the best of our knowledge this is the first time such initiative has been proposed for Sanskrit. This work represents a foundational step toward AI-enabled access to one of humanity's most significant linguistic and cultural traditions.